

Lesson 19 — Sensors Make Decisions (+ Debug)

Matches LEGO Teacher Portal lesson: "All Hands on Deck".

Objective (I can...): I can use a sensor value inside a condition so my model decides on its own. **CS**

concept: Conditions driven by sensor thresholds · **Builds on:** Lesson 18 (if/then/else).

AI Connection (2 min): AI sets a threshold too — "IF confidence is above 70%, THEN act." *First-principles question:* "What goes wrong if the cutoff is set too low?"

Materials: LEGO CS & AI kit + device per pair, objects at different distances / color cards. **Model reuse:** Conditionals model from Lesson 16 — code only.

60-minute flow - Warm-up (8): "How close is close?" — pairs guess distances and pick a cutoff (threshold) for "near" vs. "far." - **Concept + demo (10):** Demo a condition using a sensor threshold: if distance < a set value, then act; else the other branch. - **Build/Code (30):** Pairs code a sensor-threshold if/then/else (closer than X → back away; else → wait). **Debug mini-task (3 min):** give a program whose threshold is off so it never triggers — pairs find and fix the value. *Swap at 15 min.* - **Challenge (7):** Tune the threshold; test how decisions change. - **Share + pack-up (5):** Pairs show the model deciding differently at two distances. Model stays built.

Pair roles: Driver codes, Navigator moves the object and calls the threshold. *Swap at 15 min. (Trio: third leads the debug.)*

Differentiation: *Support* — provide the threshold value. *Extension / fast-finisher* — two thresholds for three outcomes. *Reteach* — watch the live distance number and pick the cutoff together.

Exit check: Pair's model changes behavior based on the sensor crossing the threshold. **At-home (optional):** Describe a real "smart" behavior (auto lights) as an if/then/else. ~10 min.
